Config

In [1]:
```python
import json
import pandas as pd
import numpy as np
import os
```

In [2]:
```python
import spacy
nlp_spacy = spacy.load("en_core_web_sm")

import nltk
nltk.download('punkt')
nltk.download('stopwords')
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords, wordnet
from nltk.stem import WordNetLemmatizer
import string
import re
```

```
[nltk_data] Downloading package punkt to
[nltk_data]     /Users/gabrielvictorgomesferreira/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data]     /Users/gabrielvictorgomesferreira/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
```

In [3]:
```python
import matplotlib.pyplot as plt
from wordcloud import WordCloud
```

In [4]:
```python
import gensim
from gensim.models.phrases import Phraser, Phrases
```

In [5]:
```python
from sklearn.feature_extraction.text import TfidfVectorizer
```

In [6]:
```python
from rake_nltk import Rake
```

In [7]:
```python
# Load the config file
with open('../config/config.json', 'r') as f:
    config = json.load(f)

file_path = config["data_loc"]
```

Datasets

Trait Phrases Dataset

In [48]:
```python
# Read trait data
file_name = "Trait dictionary.txt"
final_path = os.path.join(file_path, file_name)

# Read trait dictionary file
with open(final_path, "r", encoding="utf-8") as file:
    trait_phrases = set([phrase.lower().strip() for phrase in file if phrase.strip()])

print(f"Number of good phrases in the trait dictionary: {len(trait_phrases)}")
# print(trait_phrases)
```

```
Number of good phrases in the trait dictionary: 22719
```

Corpus Dataset

In [9]:
```python
file_name = "QTL_text.json"
final_path = os.path.join(file_path, file_name)

# Load json file
df = pd.read_json(final_path)
print(f"Shape of the original dataset: {df.shape}", "\n")

df.head()
```

```
Shape of the original dataset: (11278, 5)
```

Out[9]:

	PMID	Journal	Title	Abstract	Category
0	17179536	J Anim Sci. 2007 Jan;85(1):22-30.	Variance component analysis of quantitative tr...	In a previous study, QTL for carcass compositi...	1
1	17177700	J Anim Breed Genet. 2006 Dec;123(6):414-8.	Single nucleotide polymorphism identification,...	Pituitary adenylate cyclase-activating polypep...	0
2	17129674	Vet Parasitol. 2007 Apr 10;145(1-2):2-10. Epub...	Genetic resistance to Sarcocystis miescheriana...	Clinical and parasitological traits of Sarcocy...	0
3	17121599	Anim Genet. 2006 Dec;37(6):543-53.	Results of a whole-genome quantitative trait l...	A whole-genome quantitative trait locus (QTL) ...	1
4	17057239	Genetics. 2006 Dec;174(4):2119-27. Epub 2006 O...	Unexpected high polymorphism at the FABP4 gene...	Fatty acid bing protein 4 (FABP4) plays a key ...	0

Pre-processing and other requirements

Given Pre-Processing Requirements

In [10]:
```python
# "In this project, you will need to use "Abstract" and "Category", and you can ignore the other fields."
df = df[['Abstract', 'Category']]
print(f"Shape before filtering: {df.shape}", "\n")

# "In this project, you need to ignore papers in Category ‘0’."
df = df[df['Category'] == 1].reset_index(drop=True)
print(f"Shape after filtering: {df.shape}", "\n")

df.head()
```

```
Shape before filtering: (11278, 2)

Shape after filtering: (1007, 2)
```

Out[10]:

	Abstract	Category
0	In a previous study, QTL for carcass compositi...	1
1	A whole-genome quantitative trait locus (QTL) ...	1
2	A partial genome scan using microsatellite mar...	1
3	BACKGROUND: The rate of pubertal development a...	1
4	Previously, quantitative trait loci (QTL) for ...	1

NLP Pre-Processing

In [11]:
```python
# Set stop words
stop_words = set(stopwords.words('english'))

# Set lemmatizer
lemmatizer = WordNetLemmatizer()

# Defining NLP Pre-Processing steps. These steps were the one that highlighted the most on my exploration phase
def nlp_preprocessing(abstract_tokenized):
    abstract_tokenized = abstract_tokenized.apply(lambda doc: doc.lower()) # Lowercasing txt
    abstract_tokenized = abstract_tokenized.apply(lambda doc: doc.translate(str.maketrans("", "", string.punctuation))) # Removing punctuations from the text
    # abstract_tokenized = abstract_tokenized.apply(lambda doc: re.sub(r"\d+", "", doc)) # Removing numbers from the text
    abstract_tokenized = abstract_tokenized.apply(lambda doc: [token for token in word_tokenize(doc) if token not in stop_words]) # Tokenizing and removing stop words from the te
    abstract_tokenized = abstract_tokenized.apply(lambda doc: [lemmatizer.lemmatize(token, pos=wordnet.NOUN) for token in doc]) # Convert words to their base
    abstract_tokenized = abstract_tokenized.apply(lambda doc: [token.strip() for token in doc if token.strip() and len(token)>1]) # Removing extra space
    return abstract_tokenized

# https://medium.com/@lingostat/tf-idf-for-text-preprocessing-in-machine-learning-a66b29774040
```

In [12]:
```python
# Applying pre-defined pre-processing steps
df_processed = df.copy()
df_processed['abstract_nltk_token'] = nlp_preprocessing(df_processed['Abstract'])
df_processed['abstract_nltk_text'] = df_processed['abstract_nltk_token'].apply(lambda token: " ".join(token))
df_processed.head()
```

Out[12]:

	Abstract	Category	abstract_nltk_token	abstract_nltk_text
0	In a previous study, QTL for carcass compositi...	1	[previous, study, qtl, carcass, composition, m...	previous study qtl carcass composition meat qu...
1	A whole-genome quantitative trait locus (QTL) ...	1	[wholegenome, quantitative, trait, locus, qtl,...	wholegenome quantitative trait locus qtl scan ...
2	A partial genome scan using microsatellite mar...	1	[partial, genome, scan, using, microsatellite,...	partial genome scan using microsatellite marke...
3	BACKGROUND: The rate of pubertal development a...	1	[background, rate, pubertal, development, wean...	background rate pubertal development weaning e...
4	Previously, quantitative trait loci (QTL) for ...	1	[previously, quantitative, trait, locus, qtl, ...	previously quantitative trait locus qtl tenthr...

Analysis

Analysis Without Key-Phrases

Task 1

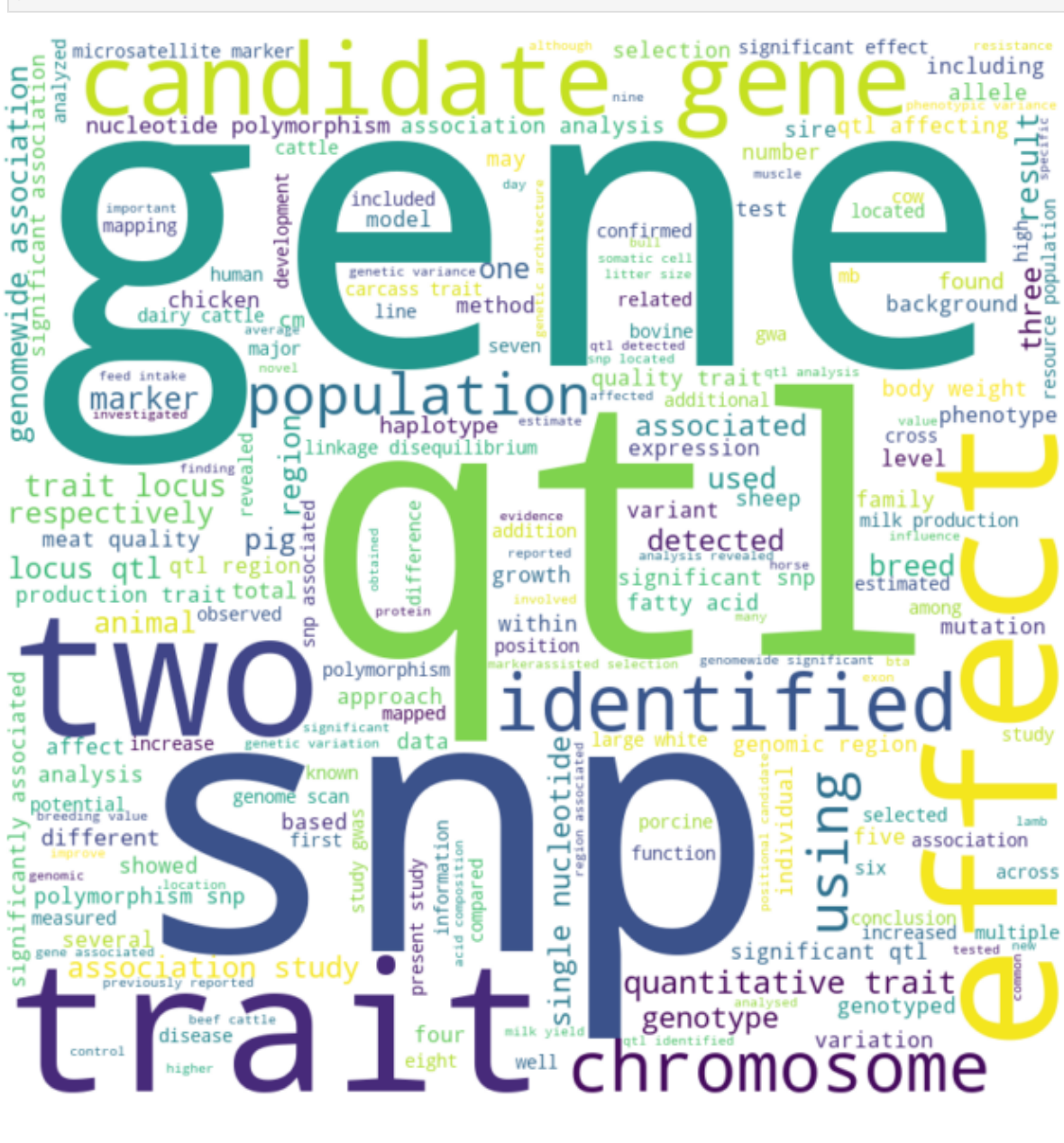
Use wordcloud to visualize words in this corpus. The figure should be 800*800, with white background color. You will need to generate two word cloud images: 1) use word frequency, and 2) use tf-idf You may find the following libraries useful for this task: WordCloud

1) Frequency Base

In [49]:
```python
# Define input text
input_text = " ".join(df_processed['abstract_nltk_text'])
# input_text
```

In [14]:
```python
# Define word cloud
wordcloud = WordCloud(width=800, height=800, background_color='white').generate(input_text)

# Display
plt.figure(figsize=(8,8))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis("off")
plt.show()
```



2) TF-IDF Base

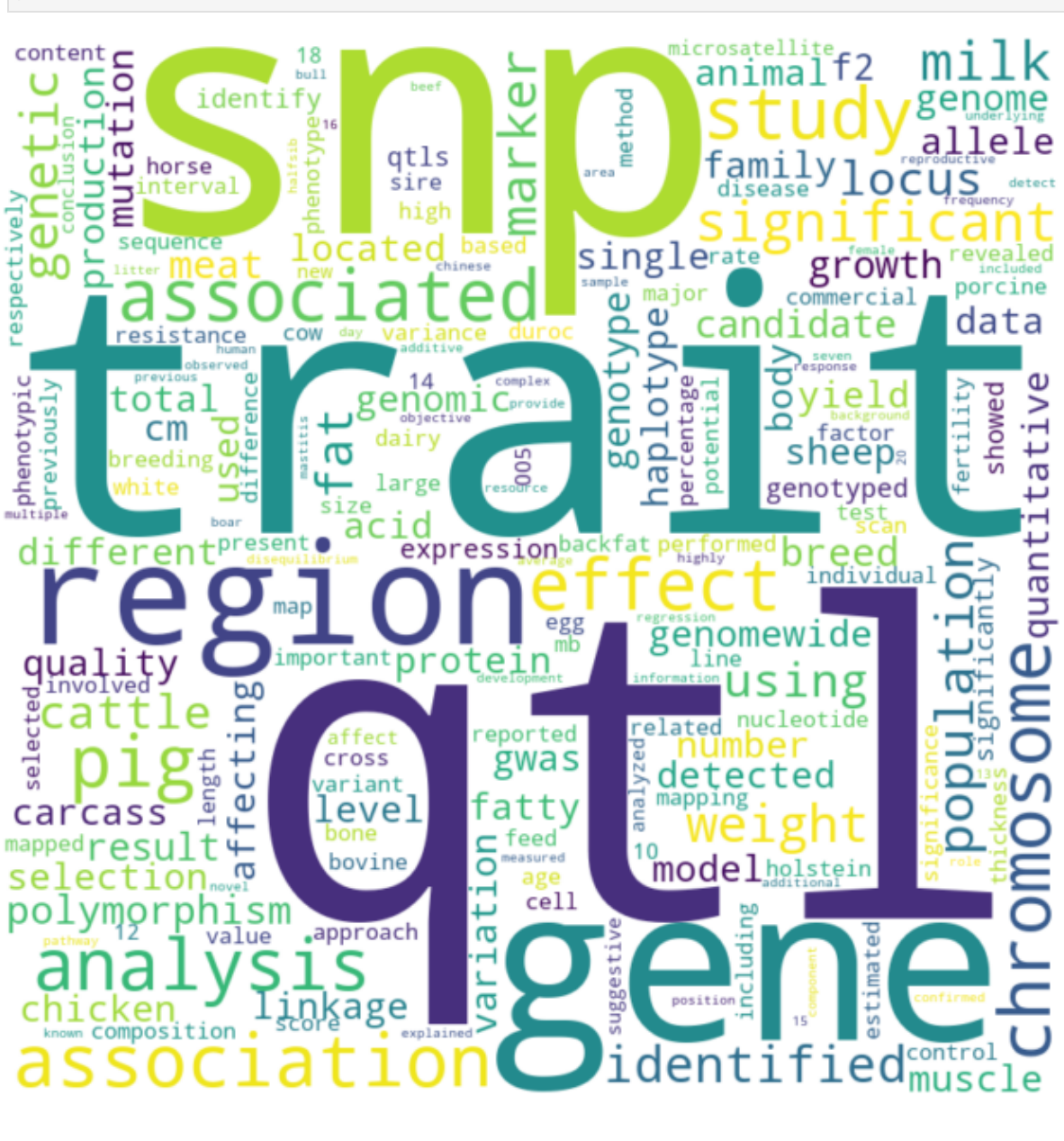
In [50]:
```python
# Define input token
input_string = df_processed['abstract_nltk_token'].apply(lambda token: " ".join(token))
# input_string
```

In [51]:
```python
# https://medium.com/@lingostat/tf-idf-for-text-preprocessing-in-machine-learning-a66b29774040

# Compute TF-IDF
tfidf_vec = TfidfVectorizer(stop_words='english')
tfidf_matrix = tfidf_vec.fit_transform(input_string)
tfidf_scores = np.asarray(tfidf_matrix.mean(axis=0)).flatten()
tfidf_word = dict(zip(tfidf_vec.get_feature_names_out(), tfidf_scores))
# print(tfidf_word)
```

In [17]:
```python
# Define word cloud
wordcloud = WordCloud(width=800, height=800, background_color='white').generate_from_frequencies(tfidf_word)

# Display
plt.figure(figsize=(8,8))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis("off")
plt.show()
```



Top TF-IDF Score Words

In [18]:
```python
# Sort and print highest ranked words
top_words = sorted(tfidf_word.items(), key=lambda x: x[1], reverse=True)[:10]
top_words_list, top_scores_list = zip(*top_words)
for word, score in top_words:
    print(f"{word}: {round(score, 4)}")
```

```
qtl: 0.0613
trait: 0.0461
snp: 0.0458
gene: 0.0397
region: 0.0283
association: 0.026
associated: 0.0254
study: 0.0253
analysis: 0.0251
pig: 0.0249
```

Task 2

Train a Word2Vec model on this corpus, with the following parameters vector_size=100, window=5, min_count=10 For each of the top 10 tf-idf words, print the 20 most similar words.

In [19]:
```python
# Document body
abstract_nltk = df_processed['abstract_nltk_token']

# Define model Word2Vec model
word2vec_model = gensim.models.Word2Vec(
    vector_size=100,
    window=5,
    min_count=10)

# Build Vocabulary
word2vec_model.build_vocab(abstract_nltk)

# Train model
word2vec_model.train(abstract_nltk, total_examples=word2vec_model.corpus_count, epochs=word2vec_model.epochs)

## https://www.youtube.com/watch?v=Q2NtCcqmIww
```

Out[19]:
```
(545583, 767850)
```

In [20]:
```python
# Dictionary to save values
most_similar_dict = {}

# Print and collect vales
for word in top_words:
    similar_words = word2vec_model.wv.most_similar(word[0], topn=20)
    print(f"{word[0]} → {', '.join([w[0] for w in similar_words])}")

    most_similar_dict[word[0]] = [(str(w[0]) + ":" + str(round(w[1], 4))) for w in similar_words]

# Create data-frame
df_test = pd.DataFrame.from_dict(most_similar_dict, orient='index')
df_test.index = range(1, len(df_test) + 1)
df_test.columns = [f"Similar {i+1}" for i in range(df_test.shape[1])]
df_test.insert(0, "Top Word", top_words_list)
df_test.insert(1, "TF-IDF Score", [round(score, 4) for score in top_scores_list])

df_test
```

```
qtl → qtls, mapped, previously, detected, reported, suggestive, chromosome, locus, identified, centromeric, several, chromosomal, putative, fat1, 13, region, chromosomewide, one,
mb, harboring
trait → affecting, quantitative, locus, economically, growth, fatness, fertility, influencing, complex, underlying, quality, resistance, beef, architecture, eggshell, meat, import
ant, performance, muscling, related
snp → exon, singlenucleotide, intron, single, nucleotide, polymorphism, five, haplotype, two, revealed, three, showed, flanking, seven, six, promoter, within, sequence, one, four
gene → candidate, positional, functional, mutation, plausible, bovine, potential, region, variant, causal, involved, pathway, several, causative, novel, promising, promoter, repor
ted, coding, suggest
region → identified, located, previously, within, novel, reported, centromeric, several, mb, one, bovine, mutation, close, candidate, positional, functional, detected, chromosoma
l, mapped, putative
association → genomewide, gwas, analysis, gwa, approach, conducted, perform, performed, revealed, regression, mapping, previous, genome, test, singlenucleotide, method, highdensit
y, metaanalysis, linkage, based
associated → found, significantly, also, close, located, several, mutation, novel, promising, one, highly, qtls, receptor, vicinity, near, contains, plausible, bovine, positional,
bta6
study → objective, identify, present, detect, aim, gwas, genome, map, conducted, fine, previous, genomic, approach, perform, scan, performed, gwa, mapping, investigate, carried
analysis → regression, genomewide, disequilibrium, linkage, mapping, association, approach, test, marker, gwa, revealed, method, nonparametric, microsatellite, gwas, combined, mix
ed, interval, based, perform
pig → commercial, duroc, chicken, chinese, resource, erhualian, white, population, large, cross, meishan, rock, breed, leghorn, pietrain, boar, broiler, dongxiang, landrace, indig
enous
```

Out[20]:

	Top Word	TF-IDF Score	Similar 1	Similar 2	Similar 3	Similar 4	Similar 5	Similar 6	Similar 7	Similar 8	...	Similar 1
1	qtl	0.0613	qtls:0.9731	mapped:0.9354	previously:0.9258	detected:0.9116	reported:0.8957	suggestive:0.8833	chromosome:0.8759	locus:0.8682	...	several:0.852
2	trait	0.0461	affecting:0.9154	quantitative:0.8997	locus:0.8818	economically:0.8612	growth:0.8589	fatness:0.8543	fertility:0.8465	influencing:0.8257	...	quality:0.803
3	snp	0.0458	exon:0.9327	singlenucleotide:0.9262	intron:0.9239	single:0.9164	nucleotide:0.9126	polymorphism:0.8972	five:0.8924	haplotype:0.8914	...	three:0.879
4	gene	0.0397	candidate:0.9694	positional:0.9394	functional:0.8922	mutation:0.8895	plausible:0.859	bovine:0.854	potential:0.8535	region:0.8365	...	involved:0.833
5	region	0.0283	identified:0.9672	located:0.9366	previously:0.9366	within:0.9211	novel:0.9154	reported:0.9035	centromeric:0.8991	several:0.8961	...	bovine:0.880
6	association	0.0260	genomewide:0.9548	gwas:0.9433	analysis:0.9336	gwa:0.8852	approach:0.8804	conducted:0.872	perform:0.8708	performed:0.87	...	mapping:0.852
7	associated	0.0254	found:0.8948	significantly:0.8939	also:0.8938	close:0.8373	located:0.8366	several:0.8281	mutation:0.827	novel:0.8227	...	highly:0.81
8	study	0.0253	objective:0.9362	identify:0.9267	present:0.9204	detect:0.919	aim:0.9143	gwas:0.8822	genome:0.8804	map:0.8709	...	previous:0.854
9	analysis	0.0251	regression:0.953	genomewide:0.9527	disequilibrium:0.9464	linkage:0.946	mapping:0.9371	association:0.9336	approach:0.924	test:0.9165	...	revealed:0.911
10	pig	0.0249	commercial:0.9654	duroc:0.9434	chicken:0.9366	chinese:0.9353	resource:0.9307	erhualian:0.9234	white:0.9226	population:0.9212	...	meishan:0.91

10 rows × 22 columns

Analysis With Key-Phrases

Task 3

Extract phrases and repeat task 1 and 2. You can be creative in phrase extraction.

Key Phrase Extractor - Gensim

In [21]:
```python
# https://tedboy.github.io/nlps/generated/generated/gensim.models.Phrases.html

# Document Body
abstract_tokenized = df_processed['abstract_nltk_token']
```

In [52]:
```python
# Bigram and trigram
bigram = Phraser(Phrases(abstract_tokenized, min_count=2, threshold=15))
trigram = Phraser(Phrases(bigram[abstract_tokenized], min_count=2, threshold=15))
bigram_token = [bigram[doc] for doc in abstract_tokenized]
trigram_token = [trigram[bigram[doc]] for doc in abstract_tokenized]
bigram_text = " ".join([" ".join(token) for token in bigram_token])
trigram_text = [" ".join(token) for token in trigram_token]
print(bigram_token[0])
# print(bigram_text)
```

```
['previous_study', 'qtl', 'carcass_composition', 'meat_quality', 'identified', 'commercial', 'finisher', 'cross', 'main_objective', 'current_study', 'confirm', 'fine_map', 'qtl',
'ssc4', 'ssc11', 'genotyping', 'increased', 'number', 'individual', 'marker', 'analyze_data', 'using', 'combined_linkage', 'linkage_disequilibrium', 'analysis', 'method', 'modifie
d', 'version', 'method', 'excludes', 'linkage_disequilibrium', 'information', 'analysis', 'enabling', 'comparison', 'result', 'based', 'linkage', 'information', 'result', 'based',
'combined_linkage', 'linkage_disequilibrium', 'information', 'nine', 'additional', 'paternal_halfsib', 'family', 'genotyped', '18', 'marker', 'resulting', 'total', '1855', 'animal
_genotyped', '15', '13', 'marker', 'ssc4', 'ssc11', 'respectively', 'qtl_affecting', 'meat_color', 'ssc4', 'confirmed', 'whereas', 'qtl_affecting', 'lm', 'weight', 'could_confirme
d', 'combined_linkage', 'linkage_disequilibrium', 'analysis', 'resulted', 'identification', 'new', 'significant', 'effect', '14', 'trait', 'chromosome', 'heritabilities', 'qtl', '
effect', 'ranged', '18', '132', 'analysis', 'contributed', 'accurate', 'positioning', 'qtl', 'characterized', 'phenotypic', 'effect', 'however', 'result', 'showed', 'even', 'great
er', 'marker_density', 'required', 'take', 'full', 'advantage', 'linkage_disequilibrium', 'information', 'identify', 'haplotype', 'associated', 'favorable', 'qtl', 'allele']
```

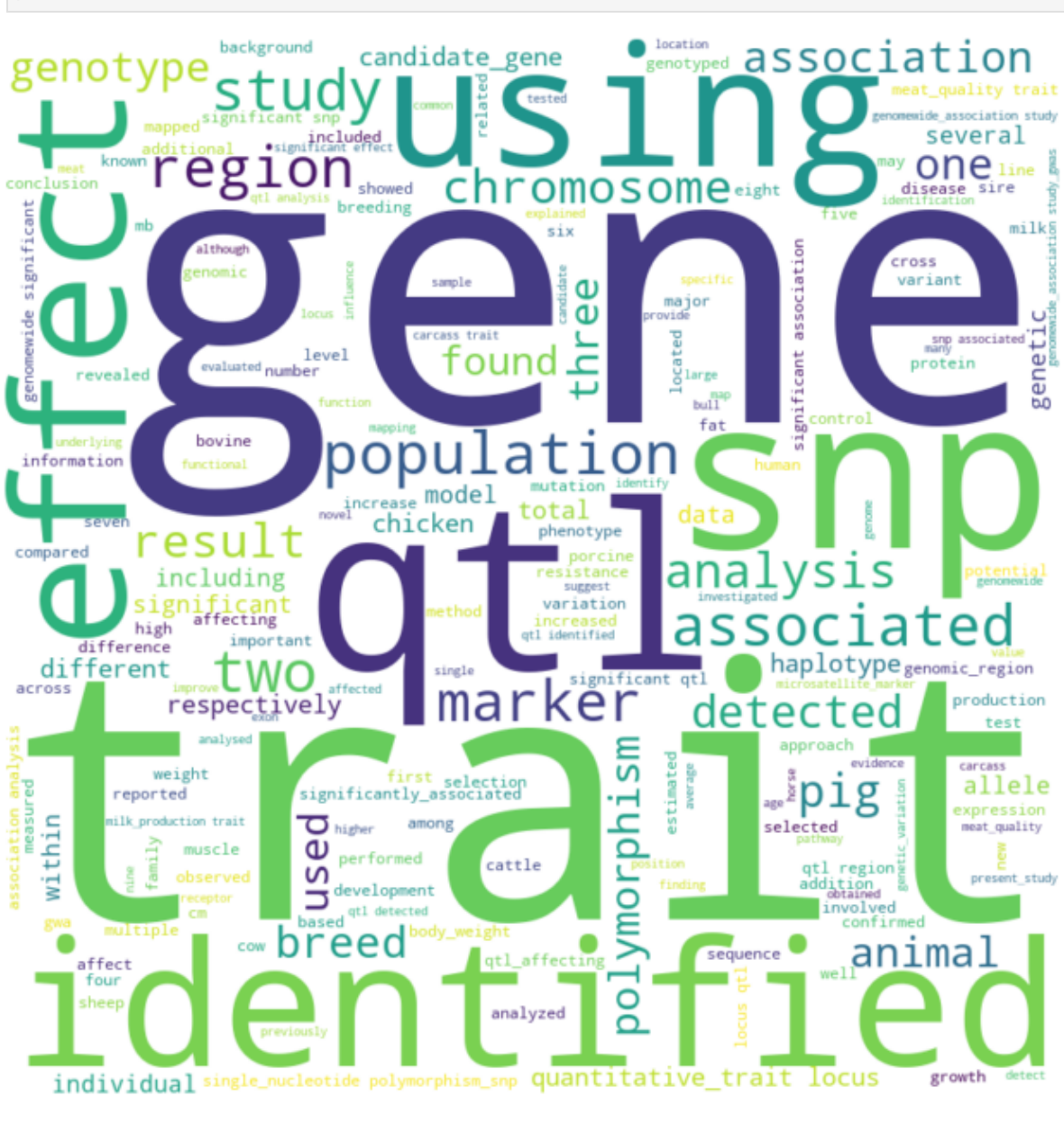
Task 3 -> Task 1 (Gensim)

Use wordcloud to visualize words in this corpus. The figure should be 800*800, with white background color. You will need to generate two word cloud images: 1) use word frequency, and 2) use tf-idf You may find the following libraries useful for this task: WordCloud

1) Frequency Based

In [23]:
```python
# Define word cloud
wordcloud = WordCloud(width=800, height=800, background_color='white').generate(bigram_text)

# Display
plt.figure(figsize=(8,8))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis("off")
plt.show()
```



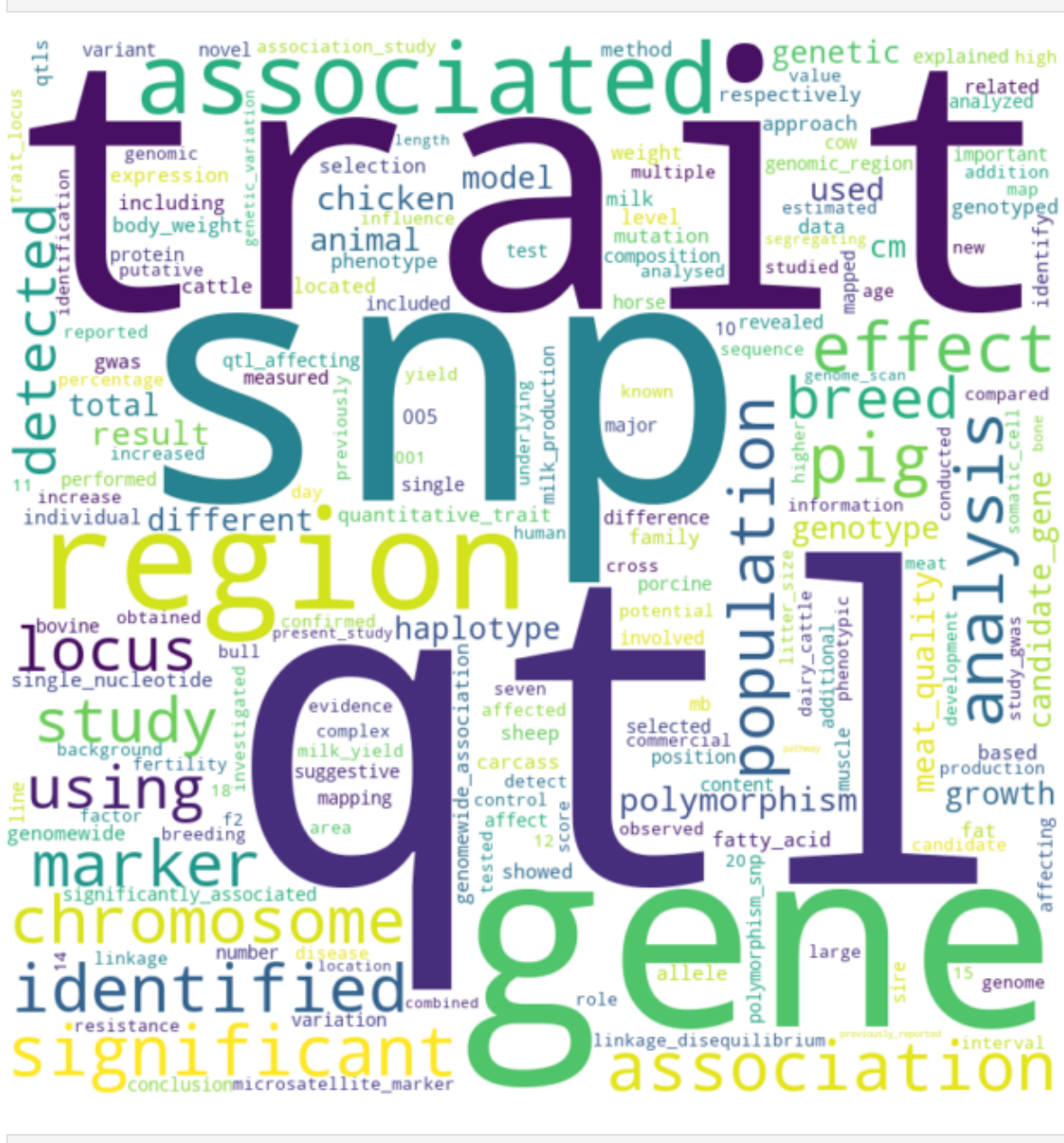
2) TF-IDF Score Base

In [24]:
```python
# Compute TF-IDF
tfidf_vec = TfidfVectorizer(stop_words='english')
tfidf_matrix = tfidf_vec.fit_transform([" ".join(token) for token in bigram_token])

tfidf_scores = np.asarray(tfidf_matrix.mean(axis=0)).flatten()
tfidf_word = dict(zip(tfidf_vec.get_feature_names_out(), tfidf_scores))

# Define word cloud
wordcloud = WordCloud(width=800, height=800, background_color='white').generate_from_frequencies(tfidf_word)

# Display
plt.figure(figsize=(8,8))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis("off")
plt.show()
```



In [25]:
```python
# Sort highest ranked words
top_words = sorted(tfidf_word.items(), key=lambda x: x[1], reverse=True)[:10]
top_words_list, top_scores_list = zip(*top_words)

for word, score in top_words:
    print(f"{word}: {round(score, 4)}")
```

```
qtl: 0.0597
trait: 0.0435
snp: 0.042
gene: 0.034
region: 0.0244
associated: 0.0232
significant: 0.023
association: 0.0225
pig: 0.0221
effect: 0.0221
```

Task 3 -> Task 2 (Gensim)

Train a Word2Vec model on this corpus, with the following parameters vector_size=100, window=5, min_count=10 For each of the top 10 tf-idf words, print the 20 most similar words.

In [26]:
```python
# Define model Word2Vec model
word2vec_model = gensim.models.Word2Vec(
    vector_size=100,
    window=5,
    min_count=10)

# Build Vocabulary
word2vec_model.build_vocab(bigram_token)

# Train model
word2vec_model.train(bigram_token, total_examples=word2vec_model.corpus_count, epochs=word2vec_model.epochs)
```

Out[26]:
```
(404682, 648925)
```

In [27]:
```python
# Dictionary to save values
most_similar_dict = {}

# Print and collect vales
for word in top_words:
    similar_words = word2vec_model.wv.most_similar(word[0], topn=20)
    print(f"{word[0]} → {', '.join([w[0] for w in similar_words])}")

    most_similar_dict[word[0]] = [(str(w[0]) + ":" + str(round(w[1], 4))) for w in similar_words]

# Create data-frame
df_test = pd.DataFrame.from_dict(most_similar_dict, orient='index')
df_test.index = range(1, len(df_test) + 1)
df_test.columns = [f"Similar {i+1}" for i in range(df_test.shape[1])]
df_test.insert(0, "Top Word", top_words_list)
df_test.insert(1, "TF-IDF Score", [round(score, 4) for score in top_scores_list])

df_test
```

```
qtl → chromosome, detected, suggestive, qtls, region, located, mb, genomewide, significant, one, identified, found, mapped, four, level, respectively, 12, 10, nine, explained
trait → growth, carcass, meat_quality, important, affect, meat, production, may, related, complex, improve, background, meat_production, fat, economically_important, genetic_archi
tecture, performance, increase, underlying, muscle
snp → genomewide, found, detected, one, significantly_associated, respectively, two, suggestive, five, located, identified, four, significant, mb, 10, three, 005, haplotype, assoc
iation, level
gene → candidate_gene, protein, positional_candidate, porcine, bovine, involved, variant, novel, coding, intron, encoding, receptor, known, mutation, potential, candidate, investi
gate, skeletal_muscle, pathway, sequence
region → located, mb, one, detected, identified, found, qtls, chromosome, suggestive, genomewide, four, respectively, mapped, 12, level, 10, two, associated, qtl, several
associated → several, novel, previously_reported, located_within, ssc8, candidate_gene, putative, genomic_region, close, highly_significant, ssc2, revealed, chromosome_10, bta6, a
lso, within, confirmed, cluster, ssc12, nine
significant → chromosome, located, genomewide, detected, suggestive, snp, found, mb, identified, one, region, qtl, respectively, qtls, four, two, level, 005, 10, significantly_ass
ociated
association → haplotype, three, five, total, analysis, seven, marker, showed, six, exon, revealed, two, significantly_associated, promoter_region, intron, containing, novel, eigh
t, tested, analysis_revealed
pig → commercial, sheep, f2, dairy, breeding_program, large_white, experimental, breed, genetic_background, therefore, genetic_basis, cattle, resource_population, breeding, litter
_size, condition, association_study, cross, architecture, collected
effect → 001, showed_significant, also, additional, threshold, revealed, 24, showed, significance, analysis_revealed, genetic_variance, observed, level, 18, phenotypic_variance, r
fi, ssc8, 0001, putative, window
```

Out[27]:

	Top Word	TF-IDF Score	Similar 1	Similar 2	Similar 3	Similar 4	Similar 5	Similar 6	Similar 7	
1	qtl	0.0597	chromosome:0.9938	detected:0.991	suggestive:0.9907	qtls:0.9905	region:0.9902	located:0.9893	mb:0.9888	genome
2	trait	0.0435	growth:0.9985	carcass:0.9973	meat_quality:0.9973	important:0.9971	affect:0.9968	meat:0.9968	production:0.9968	
3	snp	0.0420	genomewide:0.9951	found:0.9947	detected:0.9931	one:0.9929	significantly_associated:0.9927	respectively:0.9924	two:0.9923	sugge
4	gene	0.0340	candidate_gene:0.996	protein:0.994	positional_candidate:0.9939	porcine:0.9939	bovine:0.9936	involved:0.9934	variant:0.9934	n
5	region	0.0244	located:0.9975	mb:0.9974	one:0.9966	detected:0.9957	identified:0.9955	found:0.9951	qtls:0.9949	chromos
6	associated	0.0232	several:0.9982	novel:0.9981	previously_reported:0.998	located_within:0.9979	ssc8:0.9976	candidate_gene:0.9975	putative:0.9974	genomic_re
7	significant	0.0230	chromosome:0.9953	located:0.9938	genomewide:0.9936	detected:0.9931	suggestive:0.9912	snp:0.9907	found:0.9903	
8	association	0.0225	haplotype:0.9954	three:0.9951	five:0.9951	total:0.995	analysis:0.9948	seven:0.9948	marker:0.9946	sho
9	pig	0.0221	commercial:0.9988	sheep:0.9987	f2:0.9985	dairy:0.9983	breeding_program:0.9983	large_white:0.9983	experimental:0.9983	b
10	effect	0.0221	001:0.9982	showed_significant:0.9982	also:0.9981	additional:0.9981	threshold:0.9979	revealed:0.9978	24:0.9978	sho

10 rows × 22 columns

Main result (Gensim):

- 3. How many phrases you extracted can be found in the trait dictionary (by exact string matching)?

In [53]:
```python
print(f"Number of good phrases in the trait dictionary: {len(trait_phrases)}")
# print(trait_phrases)
```

```
Number of good phrases in the trait dictionary: 22719
```

In [54]:
```python
bi_unique = set()
# bi_good_phrases = [[token for token in doc if "_" in token] for doc in bigram_token]
bi_good_phrases = [[token.replace("_", " ") for token in doc] for doc in bigram_token]

for doc in bi_good_phrases:
    bi_unique.update(doc)

print(f"Number of good phrases in the trait dictionary: {len(bi_unique)}")
# print(bi_unique)
```

```
Number of good phrases in the trait dictionary: 16010
```

Analysis

In [30]:
```python
# Calculate key phrase coverage
coverage_percentage = round(len(bi_unique) / len(trait_phrases) * 100, 2)
print(f"The key phrase extractor identified the equivalent of {coverage_percentage}% of the key phrases number present in the trait file.")
```

```
The key phrase extractor identified the equivalent of 70.47% of the key phrases number present in the trait file.
```

In [31]:
```python
# Compare gensim vs. trait phrases
matching_phrases = bi_unique & trait_phrases
match_count = len(matching_phrases)

# Calculate percentage representation
match_percentage = round((match_count / len(trait_phrases)) * 100, 2)

print(f"Total Exact Matches: {match_count}")
print(f"This represents {match_percentage}% of all phrases in the trait dictionary. \n")
```

```
Total Exact Matches: 400
This represents 1.76% of all phrases in the trait dictionary.
```

In [32]:
```python
print("Summary")
print(f"Our extractor captured the equivalent of {coverage_percentage}% of the key phrases, but only {match_percentage}% were exact matches to known trait phrases.")
print("Matching Phrases:", list(matching_phrases))
```

```
Summary
Our extractor captured the equivalent of 70.47% of the key phrases, but only 1.76% were exact matches to known trait phrases.
Matching Phrases: ['conception rate', 'brain', 'fat', 'joint', 'hip', 'vascular', 'vasculogenesis', 'body length', 'acth', 'milk yield', 'hemoglobin', 'ige', 'onset', 'keel', 'cel
l', 'feather pecking', 'femoral', 't4', 'metatarsus', 'sperm motility', 'shank', 'serum leptin', 'hip width', 'uterine horn', 'fertilization', 'weight', 'ulna', 'udder depth', 'bo
dy weight', 'shank diameter', 'bmc', 'feed conversion', 'longissimus thoracis', 'jowl', 'organ', 'ca', 'phagocytosis', 'response', 'comb mass', 'length', 'igg1', 'growth', 'dopami
nergic', 'gastric', 'concentration', 'percentage', 'basophil', 'thickness', 'ileum', 'egg production', 'count', 'functional teat', 'mating', 'locomotion', 'gestation length', 'bas
eline', 'albumin', 'longissimus muscle', 'stearic acid', 'width', 'trait', 'ear size', 'vein', 'female fertility', 'chest', 'body', 'oleic acid', 'dermis', 'carcass', 'leukocyte',
'chest depth', 'respiration', 'loin muscle', 'embryo', 'purge', 'erythrocyte', 'area', 'cytokine', 'semimembranosus', 'semen volume', 'hepatocyte', 'spermatozoon', 'jejunum', 'bac
terial', 'ovary', 'dressing', 'head', 'fertility', 'body height', 'uterus', 'facial', 'hematopoiesis', 'jaw', 'lymphoma', 'energy metabolism', 'oviduct', 'yield', 'vertebra', 'bod
y conformation', 'hdl', 'clinical', 'production', 'breeding', 'oocyte', 'meiosis', 'measurement', 'spine', 'height', 'stomach', 'temperament', 'stillbirth', 'rib eye', 'heart', 't
riacylglycerol', 'gamete', 'meat ph', 'energy balance', 'oligodendrocyte', 'infection', 'tarsometatarsus', 'leg', 'artery', 'ability', 'tail', 'thermoregulation', 'rump', 'novelt
y', 'gizzard', 'diameter', 'egg weight', 'neurotransmitter', 'platelet', 'hindquarter', 'milk urea', 'somite', 'loin', 'macrophage', 'feather', 'rdw', 'pelvis', 'appendage', 'horn
length', 'monocyte', 'neck', 'number stillborn', 'mortality', 'iga', 'back fat', 'cocaine', 'content', 'perineum', 'lifespan', 'forelimb', 'embryonic development', 'vertebrate', '
connective tissue', 'morphology', 'endometrium', 'fatty acid', 'behavior', 'fat thickness', 'ejaculation', 'litter', 'serum ige', 'wound healing', 'epidermis', 'eye muscle', 'ham
weight', 'hindlimb', 'heart weight', 'adipose', 'small intestine', 'mchc', 'male fertility', 'osteoclast', 'pericardium', 'splenic', 'sensory', 'semen', 'corticotropin', 'palmitol
eic acid', 'nipple number', 'ovulation rate', 'radius', 'haemoglobin', 'rumen', 'bacteria', 'bone', 'mch', 'semimembranosus muscle', 'bw', 'ribeye', 'duodenum', 'udder morpholog
y', 'antibody', 'digestion', 'hair', 'loin eye', 'bone strength', 'vision', 'eyelash', 'belly', 'heat', 'angiogenesis', 'thoracic vertebra', 'lymphocyte', 'blood', 'homeostasis',
'teat number', 'gizzard weight', 'cornea', 'nipple', 'neutrophil', 'metabolism', 'ossification', 'reproduction', 'hypothalamus', 'nervous system', 'interleukin', 'drumstick', 'ske
letal muscle', 'sperm', 'failure', 'litter size', 'gestation', 'number', 'immune system', 'shank length', 'humerus', 'adipocyte', 'bone density', 'postnatal growth', 'placenta', '
ce', 'rearing', 'backfat', 'subcutaneous fat', 'scrotal circumference', 'linoleic acid', 'mcv', 'spermatogenesis', 'eosinophil', 'uterine', 'ige level', 'abdominal fat', 'triglyce
ride', 'egg', 'leg muscle', 'chemokine', 'serum lipid', 'skin', 'lipid', 'milk', 'intramuscular fat', 'meat', 'epididymis', 'kidney', 'chest circumference', 'rbc', 'muscle area',
'craniofacial', 'gluconeogenesis', 'consumption', 'meat tenderness', 'heart girth', 'sarcomere', 'adrenal gland', 'feathering', 'antibody response', 'survival', 'cortical bone', '
infarction', 'offspring', 'carcass weight', 'abomasum', 'plateletcrit', 'parasite', 'bmd', 'lung', 'igg', 'cholesterol', 'liver', 'waist', 'pancreas', 'capillary', 'carcass lengt
h', 'cartilage', 'ra', 'cns', 'adipose tissue', 'gland', 'alpha', 'interferon', 'femur', 'ligament', 'maternal behavior', 'comb', 'lactation', 'intestine', 'organism', 'cerebellu
m', 'amphetamine', 'iris', 'ear', 'pregnancy rate', 'balance', 'teat', 'testis', 'backfat thickness', 'spleen', 'stroke', 'mineral', 'fleece', 'hemoglobin concentration', 'tibia',
'bone mineral', 'calcium', 'chest width', 'fetal growth', 'breast muscle', 'fillet', 'amount', 'strength', 'wing', 'sclerotome', 'lipid metabolism', 'ham', 'thymocyte', 'tender',
'skin thickness', 'dressing percentage', 'meat color', 'vertebral', 'urine', 'limb', 'body size', 'viability', 'viable', 'level', 'ovulation', 'lumbar vertebra', 'cardiovascular',
'time', 'social behavior', 'blood cell', 'immune response', 'proventriculus', 'mammary gland', 'wool', 'thyroxine', 'vasoconstriction', 'colostrum', 'thymus', 'ribeye area', 'ri
b', 'hct', 'vocalization', 'ldl', 'synapse', 'osteogenesis', 'feeding behavior', 'serotonin', 'immunoglobulin', 'glycolytic potential', 'longissimus dorsi', 'sterol', 'eye', 'udde
r', 'marbling', 'thigh', 'tlr2', 'birth', 'muscle fiber', 'shoulder', 'protein', 'tumor', 'catecholamine', 'eyelid', 'muscle', 'glucagon', 'ap', 'triglyceride level', 'chest girt
h', 'iron', 'ovarian follicle', 'tibia length', 'lactose yield', 'milk protein', 'abnormal sperm', 'hematocrit', 'parturition', 'dopamine', 'taste', 'leaf fat', 'myocardial', 'pla
sma cortisol', 'horn', 'breast', 'scapula', 'fat depth']
```

Key Phrase Extractor - Rake

In [33]:
```python
# Define Rake Extractor
r = Rake(stopwords=stop_words, punctuations=string.punctuation, min_length=1, max_length=3)

# Extract RAKE key phrases per document
df_processed["key_phrases_rake"] = df_processed["Abstract"].apply(
    lambda text: r.extract_keywords_from_text(text) or [phrase for score, phrase in r.get_ranked_phrases_with_scores() if score > 5]
)

# Preprocess key phrases and merge with documents
df_processed["processed_key_phrases"] = df_processed["key_phrases_rake"].apply(lambda phrases: ["_".join(nlp_preprocessing(pd.Series([phrase]))[0]) for phrase in phrases])

# Merge key phrases back into original documents
df_processed["abstract_rake_token"] = df_processed.apply(lambda row: row["abstract_nltk_token"] + row["processed_key_phrases"], axis=1)
df_processed['abstract_rake_text'] = df_processed['abstract_rake_token'].apply(lambda token: " ".join(token))
df_processed.head()
```

Out[33]:

	Abstract	Category	abstract_nltk_token	abstract_nltk_text	key_phrases_rake	processed_key_phrases	abstract_rake_token	abstract_rake_text
0	In a previous study, QTL for carcass compositi...	1	[previous, study, qtl, carcass, composition, m...	previous study qtl carcass composition meat qu...	[take full advantage, new significant effects,...	[take_full_advantage, new_significant_effect, ...	[previous, study, qtl, carcass, composition, m...	previous study qtl carcass composition meat qu...
1	A whole-genome quantitative trait locus (QTL) ...	1	[wholegenome, quantitative, trait, locus, qtl,...	wholegenome quantitative trait locus qtl scan ...	[entire porcine genome, based nonparametric ap...	[entire_porcine_genome, based_nonparametric_ap...	[wholegenome, quantitative, trait, locus, qtl,...	wholegenome quantitative trait locus qtl scan ...
2	A partial genome scan using microsatellite mar...	1	[partial, genome, scan, using, microsatellite,...	partial genome scan using microsatellite marke...	[porcine resource population, marker loci sw13...	[porcine_resource_population, marker_locus_sw1...	[partial, genome, scan, using, microsatellite,...	partial genome scan using microsatellite marke...
3	BACKGROUND: The rate of pubertal development a...	1	[background, rate, pubertal, development, wean...	background rate pubertal development weaning e...	[usmarc resource population, specific hydroxys...	[usmarc_resource_population, specific_hydroxys...	[background, rate, pubertal, development, wean...	background rate pubertal development weaning e...
4	Previously, quantitative trait loci (QTL) for ...	1	[previously, quantitative, trait, locus, qtl, ...	previously quantitative trait locus qtl tenthr...	[quantitative trait loci, loin eye area, janus...	[quantitative_trait_locus, loin_eye_area, janu...	[previously, quantitative, trait, locus, qtl, ...	previously quantitative trait locus qtl tenthr...

Task 3 -> Task 1 (Rake)

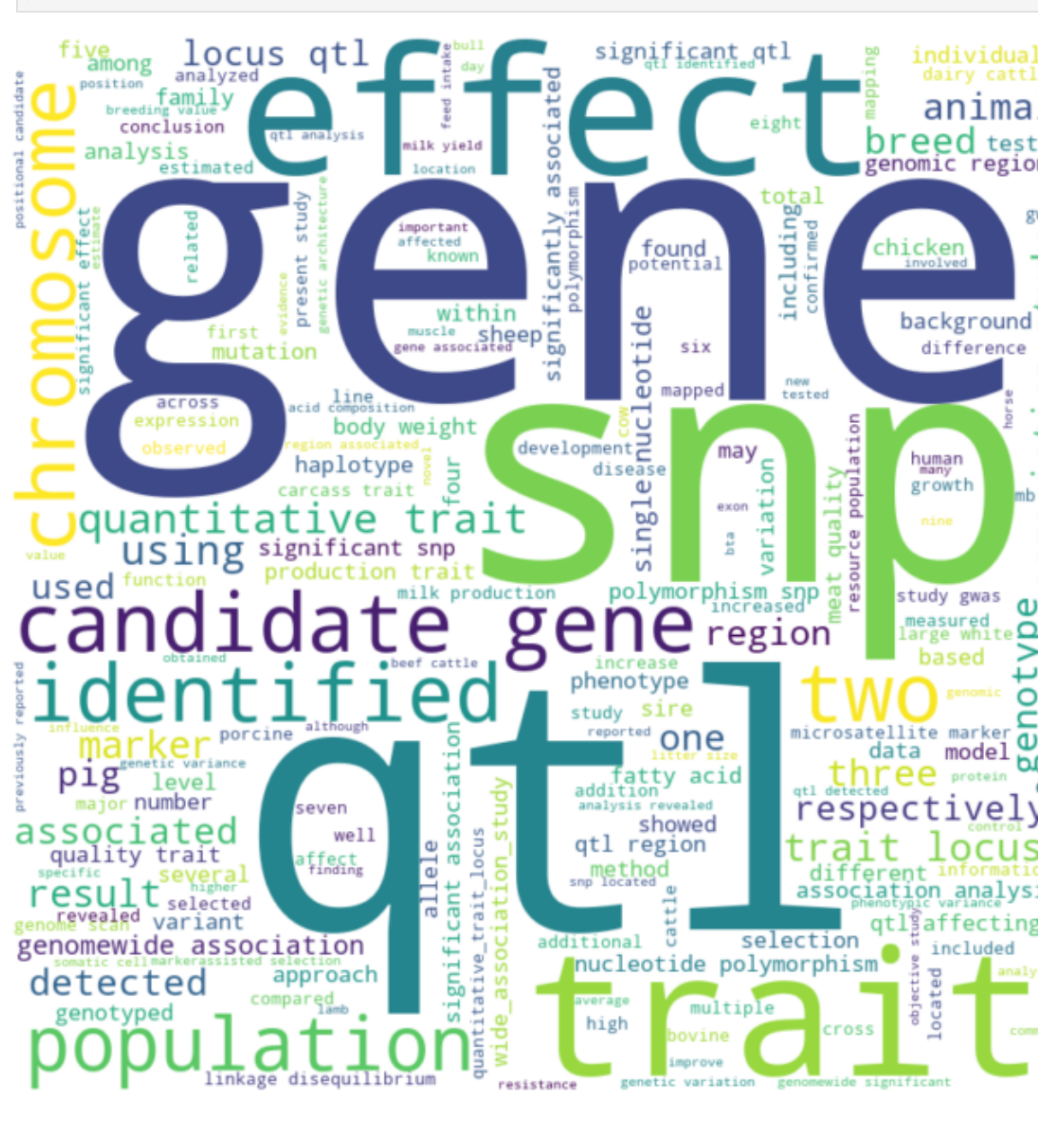
Use wordcloud to visualize words in this corpus. The figure should be 800*800, with white background color. You will need to generate two word cloud images: 1) use word frequency, and 2) use tf-idf You may find the following libraries useful for this task: WordCloud

1) Frequency Based

In [55]:
```python
input_text_rake = " ".join(df_processed['abstract_rake_text'])
# input_text_rake
```

In [35]:
```python
# Define word cloud
wordcloud = WordCloud(width=800, height=800, background_color='white').generate(input_text_rake)

# Display
plt.figure(figsize=(8,8))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis("off")
plt.show()
```



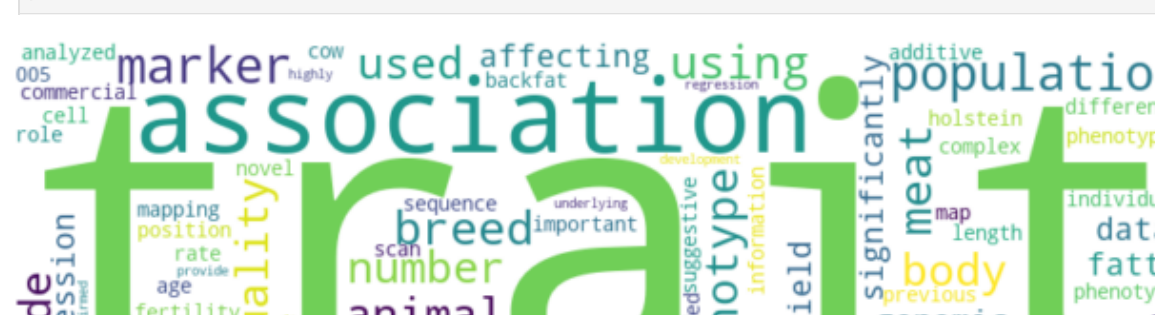
2) TF-IDF Score Base

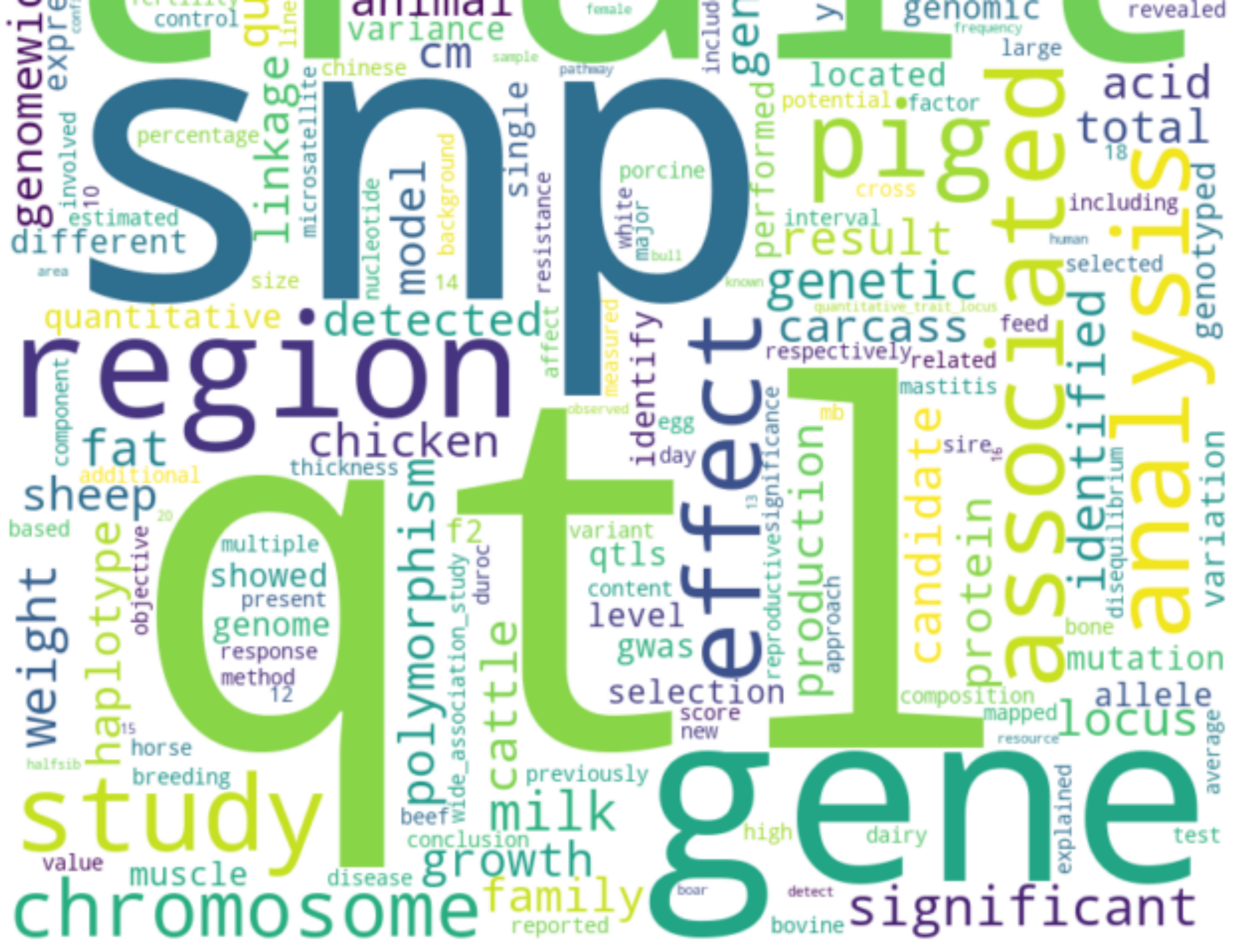
In [56]:
```python
# Define input token
input_string = df_processed['abstract_rake_token'].apply(lambda token: " ".join(token))
# input_string
```

In [57]:
```python
# Compute TF-IDF
tfidf_vec = TfidfVectorizer(stop_words='english')
tfidf_matrix = tfidf_vec.fit_transform(input_string)
tfidf_scores = np.asarray(tfidf_matrix.mean(axis=0)).flatten()
tfidf_word = dict(zip(tfidf_vec.get_feature_names_out(), tfidf_scores))
# print(tfidf_word)
```

In [38]:
```python
# Define word cloud
wordcloud = WordCloud(width=800, height=800, background_color='white').generate_from_frequencies(tfidf_word)

# Display
plt.figure(figsize=(8,8))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis("off")
plt.show()
```





Top TF-IDF Score Words

In [39]:
```
# Sort and print highest ranked words
top_words = sorted(tfidf_word.items(), key=lambda x: x[1], reverse=True)[:10]
top_words_list, top_scores_list = zip(*top_words)
for word, score in top_words:
    print(f"{word}: {round(score, 4)}")
```

```
qtl: 0.0573
trait: 0.0432
snp: 0.0431
gene: 0.0374
region: 0.0265
association: 0.0245
associated: 0.0239
study: 0.0237
analysis: 0.0235
pig: 0.0234
```

Task 3 -> Task 2 (Rake)

Train a Word2Vec model on this corpus, with the following parameters vector_size=100, window=5, min_count=10 For each of the top 10 tf-idf words, print the 20 most similar words.

In [40]:
```
# Document body
abstract_nltk = df_processed['abstract_rake_token']

# Define model Word2Vec model
word2vec_model = gensim.models.Word2Vec(
    vector_size=100,
    window=5,
    min_count=10)

# Build Vocabulary
word2vec_model.build_vocab(abstract_nltk)

# Train model
word2vec_model.train(abstract_nltk, total_examples=word2vec_model.corpus_count, epochs=word2vec_model.epochs)
```

Out[40]:
```
(555416, 829860)
```

In [41]:
```
# Dictionary to save values
most_similar_dict = {}

# Print and collect vales
for word in top_words:
    similar_words = word2vec_model.wv.most_similar(word[0], topn=20)
    print(f"{word[0]} → {', '.join([w[0] for w in similar_words])}")

    most_similar_dict[word[0]] = [(str(w[0]) + ":" + str(round(w[1], 4))) for w in similar_words]

# Create data-frame
df_test = pd.DataFrame.from_dict(most_similar_dict, orient='index')
df_test.index = range(1, len(df_test) + 1)
df_test.columns = [f"Similar {i+1}" for i in range(df_test.shape[1])]
df_test.insert(0, "Top Word", top_words_list)
df_test.insert(1, "TF-IDF Score", [round(score, 4) for score in top_scores_list])

df_test
```

```
qtl → qtls, detected, mapped, previously, suggestive, reported, chromosome, 13, four, chromosomewide, region, significant, identified, mb, 14, chromosomal, locus, found, genomewis
e, several
trait → affecting, quantitative, locus, growth, fatness, meat, quality, carcass, fertility, economically, production, influencing, affect, endocrine, related, bone, eggshell, comp
lex, performance, tenderness
snp → single, exon, three, haplotype, revealed, five, singlenucleotide, intron, nucleotide, two, polymorphism, one, showed, seven, six, within, four, flanking, array, identified
gene → candidate, positional, mutation, potential, bovine, functional, involved, several, causal, pathway, plausible, porcine, variant, region, promising, reported, function, know
n, receptor, close
region → identified, located, mb, previously, within, reported, one, several, novel, bovine, chromosome, detected, putative, mutation, found, intron, chromosomal, four, near, mapp
ed
association → genomewide, gwas, analysis, gwa, conducted, performed, revealed, previous, perform, singlenucleotide, result, imputed, approach, regression, mapping, chip, study, de
tect, array, single
associated → significantly, found, also, qtls, fabp4, several, mutation, coincided, receptor, reported, close, bovine, highly, harbouring, located, promising, one, near, linked, p
romoter
study → identify, objective, aim, detect, present, gwas, conducted, previous, map, performed, perform, scan, genome, conduct, association, fine, approach, result, gwa, genomewide
analysis → regression, genomewide, mapping, disequilibrium, linkage, gwa, approach, association, conducted, gwas, method, performed, marker, perform, microsatellite, test, impute
d, combined, revealed, previous
pig → commercial, resource, duroc, chinese, rock, erhualian, sutai, breed, leghorn, cross, population, meishan, dongxiang, chicken, intercross, white, pietrain, landrace, f2, broi
ler
```

Out[41]:

	Top Word	TF-IDF Score	Similar 1	Similar 2	Similar 3	Similar 4	Similar 5	Similar 6	Similar 7	Similar 8	...	Similar 11
1	qtl	0.0573	qtls:0.9612	detected:0.9314	mapped:0.931	previously:0.9038	suggestive:0.9022	reported:0.8981	chromosome:0.8872	13:0.8621	...	region:0.8576
2	trait	0.0432	affecting:0.9189	quantitative:0.8932	locus:0.8829	growth:0.8655	fatness:0.8597	meat:0.8321	quality:0.8321	carcass:0.8148	...	production:0.7728
3	snp	0.0431	single:0.9156	exon:0.9056	three:0.901	haplotype:0.8982	revealed:0.8968	five:0.8928	singlenucleotide:0.8902	intron:0.8875	...	polymorphism:0.872
4	gene	0.0374	candidate:0.9795	positional:0.9467	mutation:0.8859	potential:0.8516	bovine:0.8495	functional:0.8473	involved:0.8446	several:0.8404	...	plausible:0.8127
5	region	0.0265	identified:0.9772	located:0.9734	mb:0.9388	previously:0.9369	within:0.925	reported:0.9138	one:0.9099	several:0.9081	...	chromosome:0.891
6	association	0.0245	genomewide:0.9567	gwas:0.9322	analysis:0.9203	gwa:0.8774	conducted:0.8764	performed:0.8717	revealed:0.8705	previous:0.8698	...	result:0.8449
7	associated	0.0239	significantly:0.9162	found:0.9109	also:0.899	qtls:0.8549	fabp4:0.8385	several:0.8343	mutation:0.8313	coincided:0.8261	...	close:0.8134
8	study	0.0237	identify:0.9315	objective:0.9314	aim:0.9176	detect:0.917	present:0.901	gwas:0.9006	conducted:0.8629	previous:0.8567	...	perform:0.8448
9	analysis	0.0235	regression:0.9603	genomewide:0.9522	mapping:0.9463	disequilibrium:0.9419	linkage:0.9386	gwa:0.9362	approach:0.9304	association:0.9203	...	method:0.9145
10	pig	0.0234	commercial:0.9636	resource:0.9628	duroc:0.9543	chinese:0.9456	rock:0.9415	erhualian:0.9355	sutai:0.9352	breed:0.9346	...	population:0.9281

10 rows × 22 columns

Main result (Rake):

- 3. How many phrases you extracted can be found in the trait dictionary (by exact string matching)?

In [58]:
```
print(f"Number of good phrases in the trait dictionary: {len(trait_phrases)}")
# print(trait_phrases)
```

```
Number of good phrases in the trait dictionary: 22719
```

In [59]:
```
bi_unique = set()
good_phrases = df_processed['abstract_rake_token']
for doc in good_phrases:
    bi_unique.update(doc)

print(f"Number of good phrases in the trait dictionary: {len(bi_unique)}")
# print(bi_unique)
```

```
Number of good phrases in the trait dictionary: 22245
```

Analysis

In [44]:
```
# Calculate key phrase coverage
coverage_percentage = round(len(bi_unique) / len(trait_phrases) * 100, 2)
print(f"The key phrase extractor identified the equivalent of {coverage_percentage}% of the key phrases number present in the trait file.")
```

```
The key phrase extractor identified the equivalent of 97.91% of the key phrases number present in the trait file.
```

In [45]:
```
# Compare gensim vs. trait phrases
matching_phrases = bi_unique & trait_phrases
match_count = len(matching_phrases)

# Calculate percentage representation
match_percentage = round((match_count / len(trait_phrases)) * 100, 2)

print(f"Total Exact Matches: {match_count}")
print(f"This represents {match_percentage}% of all phrases in the trait dictionary. \n")
```

```
Total Exact Matches: 286
This represents 1.26% of all phrases in the trait dictionary.
```

In [46]:
```
print("Summary")
print(f"Our extractor captured the equivalent of {coverage_percentage}% of the key phrases, but only {match_percentage}% were exact matches to known trait phrases.")
print("Matching Phrases:", list(matching_phrases))
```

```
Summary
Our extractor captured the equivalent of 97.91% of the key phrases, but only 1.26% were exact matches to known trait phrases.
Matching Phrases: ['organism', 'brain', 'vertebra', 'fat', 'cerebellum', 'vision', 'eyelash', 'joint', 'belly', 'hdl', 'amphetamine', 'clinical', 'hip', 'heat', 'iris', 'productio
n', 'angiogenesis', 'ear', 'vascular', 'lymphocyte', 'vasculogenesis', 'blood', 'breeding', 'homeostasis', 'balance', 'teat', 'testis', 'oocyte', 'spleen', 'meiosis', 'stroke', 'a
cth', 'cornea', 'nipple', 'measurement', 'mineral', 'hemoglobin', 'ige', 'fleece', 'neutrophil', 'metabolism', 'spine', 'tibia', 'height', 'onset', 'keel', 'calcium', 'cell', 'oss
ification', 'reproduction', 'femoral', 'stomach', 'hypothalamus', 'stillbirth', 'temperament', 'heart', 'interleukin', 't4', 'metatarsus', 'drumstick', 'triacylglycerol', 'fille
t', 'gamete', 'amount', 'shank', 'sperm', 'strength', 'wing', 'oligodendrocyte', 'sclerotome', 'fertilization', 'tarsometatarsus', 'infection', 'failure', 'ham', 'weight', 'thymoc
yte', 'tender', 'gestation', 'leg', 'number', 'artery', 'ulna', 'bmc', 'ability', 'tail', 'thermoregulation', 'humerus', 'jowl', 'organ', 'rump', 'vertebral', 'urine', 'novelty', '
gizzard', 'ca', 'phagocytosis', 'diameter', 'response', 'adipocyte', 'limb', 'length', 'viability', 'igg1', 'growth', 'viable', 'placenta', 'gastric', 'concentration', 'level', '
dopaminergic', 'ce', 'percentage', 'backfat', 'rearing', 'basophil', 'neurotransmitter', 'ovulation', 'platelet', 'thickness', 'hindquarter', 'cardiovascular', 'time', 'mcv', 'pro
ventriculus', 'spermatogenesis', 'eosinophil', 'ileum', 'somite', 'loin', 'feather', 'macrophage', 'cecum', 'wool', 'count', 'rdw', 'uterine', 'pelvis', 'appendage', 'mating', 'lo
comotion', 'thyroxine', 'baseline', 'monocyte', 'triglyceride', 'vasoconstriction', 'neck', 'egg', 'mortality', 'colostrum', 'iga', 'albumin', 'thymus', 'rib', 'hct', 'content', '
vocalization', 'cocaine', 'chemokine', 'ldl', 'width', 'skin', 'lipid', 'milk', 'trait', 'perineum', 'lifespan', 'meat', 'forelimb', 'synapse', 'osteogenesis', 'vein', 'vertebrat
e', 'morphology', 'serotonin', 'immunoglobulin', 'epididymis', 'endometrium', 'kidney', 'rbc', 'craniofacial', 'chest', 'body', 'behavior', 'ejaculation', 'dermis', 'litter', 'ste
rol', 'epidermis', 'carcass', 'eye', 'gluconeogenesis', 'consumption', 'leukocyte', 'udder', 'marbling', 'hindlimb', 'adipose', 'thigh', 'respiration', 'tlr2', 'birth', 'shoulde
r', 'mchc', 'protein', 'sarcomere', 'tumor', 'feathering', 'catecholamine', 'eyelid', 'osteoclast', 'muscle', 'glucagon', 'ap', 'embryo', 'purge', 'pericardium', 'erythrocyte', 's
urvival', 'splenic', 'area', 'infarction', 'offspring', 'sensory', 'cytokine', 'semimembranosus', 'corticotropin', 'semen', 'plateletcrit', 'abomasum', 'parasite', 'hepatocyte', '
spermatozoon', 'jejunum', 'lung', 'bmd', 'radius', 'haemoglobin', 'igg', 'cholesterol', 'liver', 'waist', 'iron', 'pancreas', 'capillary', 'bacterial', 'hematocrit', 'rumen', 'car
tilage', 'parturition', 'dopamine', 'bone', 'ovary', 'ra', 'dressing', 'cns', 'bacteria', 'gland', 'mch', 'head', 'fertility', 'taste', 'alpha', 'myocardial', 'interferon', 'bw', '
femur', 'uterus', 'facial', 'ligament', 'hematopoiesis', 'jaw', 'ribeye', 'horn', 'breast', 'lymphoma', 'comb', 'duodenum', 'antibody', 'lactation', 'intestine', 'oviduct', 'scap
ula', 'digestion', 'yield', 'hair']
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